

LentiMorpho: A Display that Switches Between View-Dependent and View-Independent Modes by Varying Liquid Volume

Rintaro Akiyama
Hosei University
Tokyo, Japan
rintaro.akiyama.3i@stu.hosei.ac.jp

Kouyou Otsu
Aoyama Gakuin University
Kanagawa, Japan
otsu.kouyou@it.aoyama.ac.jp

Mai Kamihori
Aoyama Gakuin University
Kanagawa, Japan
mai.kamihori@x-lab.team

Yuichi Itoh
Aoyama Gakuin University
Kanagawa, Japan
ito@it.aoyama.ac.jp

Takafumi Koike
Hosei University
Tokyo, Japan
takafumi@hosei.ac.jp

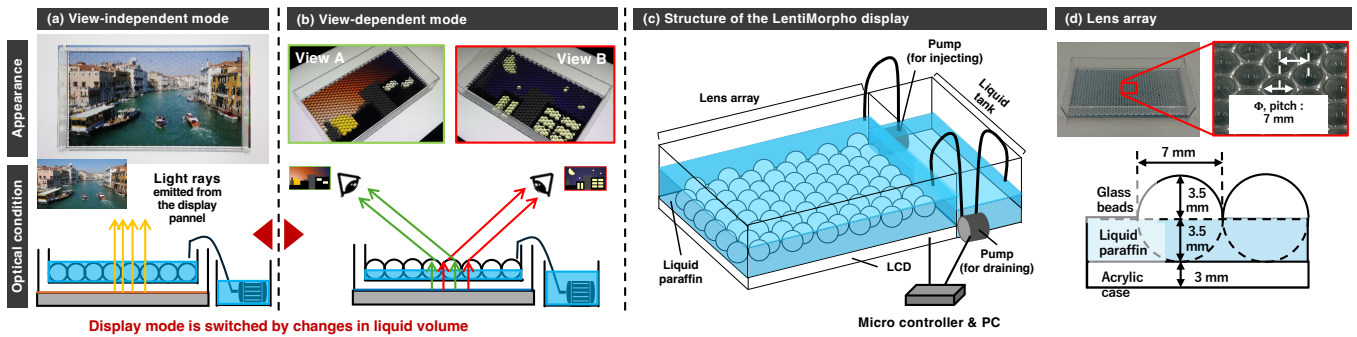


Figure 1: Overview of LentiMorpho. The display switches between view-dependent and view-independent modes by changing the optical state of the lens array through liquid volume control. In the view-dependent mode, a precomputed pattern is displayed to enable directional light separation.

Abstract

This paper presents LentiMorpho, a display system that dynamically switches between view-dependent and view-independent presentation using the same hardware. The system achieves this switching by electrically controlling the optical state of a lens array through liquid volume control. Lenses exposed to air act as refractive elements for directional light separation, whereas immersion in a liquid with a similar refractive index suppresses this refractive function and results in view-independent display behavior. Unlike conventional view-dependent displays, LentiMorpho preserves the original image quality of the underlying flat-panel display by presenting identical content uniformly across all viewing directions. LentiMorpho offers a design option for incorporating view-dependent visualization modes into conventional view-independent displays.

CCS Concepts

• **Hardware** → Emerging interfaces; • **Human-centered computing** → Displays and imagers.



This work is licensed under a Creative Commons Attribution 4.0 International License. *SIGGRAPH Emerging Technologies '26*, Los Angeles, CA, USA
© 2026 Copyright held by the owner/author(s).
ACM ISBN 979-8-4007-2549-4/26/07
<https://doi.org/10.1145/3799826.3812436>

Keywords

view-dependent displays, lens array based display, lens arrays, light-fields, optics

ACM Reference Format:

Rintaro Akiyama, Kouyou Otsu, Mai Kamihori, Yuichi Itoh, and Takafumi Koike. 2026. LentiMorpho: A Display that Switches Between View-Dependent and View-Independent Modes by Varying Liquid Volume. In *Special Interest Group on Computer Graphics and Interactive Techniques Conference Emerging Technologies (SIGGRAPH Emerging Technologies '26)*, July 19–23, 2026, Los Angeles, CA, USA. ACM, New York, NY, USA, 3 pages. <https://doi.org/10.1145/3799826.3812436>

1 Introduction

View-dependent displays present different visual content depending on the viewing direction by employing optical elements placed in front of a flat-panel display [Levoy and Hanrahan 1996] [Takaki and Nago 2010] [Takehi et al. 2005]. While this approach enables directional image separation, the optical elements remain permanently active even when view-dependent presentation is unnecessary. As a result, identical content presented uniformly across viewing directions still suffers from degraded image quality due to the view-dependent optical structure. This limitation arises not from view-dependent presentation itself, but from the inability to disable the optical elements when they are not required.

This issue becomes particularly apparent when a single display alternates between individualized and shared viewing. Although view-dependent presentation is useful for personalized content, displays are often expected to present identical content to all viewers while preserving the original image quality of the flat-panel display. In conventional systems, however, optical elements responsible for directional separation continue to affect the image even when view-dependent presentation is unnecessary, preventing the same hardware from supporting both requirements without compromising image quality.

If the optical behavior responsible for directional light separation could be selectively enabled and suppressed, a single display could support both view-dependent and view-independent presentation without such trade-offs [Lanman et al. 2010]. This capability would allow view-dependent techniques to function as a controllable mode rather than a permanently imposed constraint.

In this work, we present LentiMorpho, a display that dynamically switches between view-dependent and view-independent presentation using the same hardware. LentiMorpho preserves the original image quality of the underlying flat-panel display when presenting identical content across viewing directions, while supporting view-dependent presentation when required. By integrating both presentation modes using the same hardware, LentiMorpho provides a design option for incorporating view-dependent visualization techniques into conventional view-independent displays.

2 System Configuration

LentiMorpho is a display system that dynamically switches between view-dependent and view-independent presentation by controlling whether a lens array refracts emitted light or allows it to propagate without directional separation. This switching is achieved using the same hardware by changing the optical state of the same physical structure.

The system consists of a flat-panel display, a glass-bead lens array enclosed in a transparent container, and a liquid control mechanism that adjusts the liquid volume inside the container. Figure 1(c) shows the structure of the lens array designed in this study. The lens array is formed by arranging glass beads in a two-dimensional lattice and placing the container directly in front of the display surface. The container is partially filled with a liquid whose refractive index is similar to that of the glass beads. Changing the liquid volume alters the surrounding medium of the beads and determines whether the lens array produces refraction.

Figure 2 illustrates the schematic configuration of the lens array and the liquid control mechanism. The liquid volume is electrically controlled using separate pumps for injection and drainage, both connected to a shared external reservoir. These pumps are controlled by a microcontroller, enabling switching between presentation modes without mechanical movement of the display or optical components. As the optical state changes, the refractive effect of the lens array gradually decreases, producing a visible transition in which the lens effect appears to “dissolve,” smoothly shifting from view-dependent to view-independent display behavior. In view-dependent presentation, display content is pre-computed using ray tracing based on the geometric configuration of

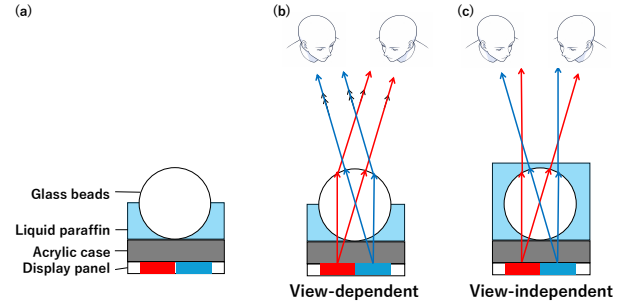


Figure 2: Lens configuration and optical path in each mode

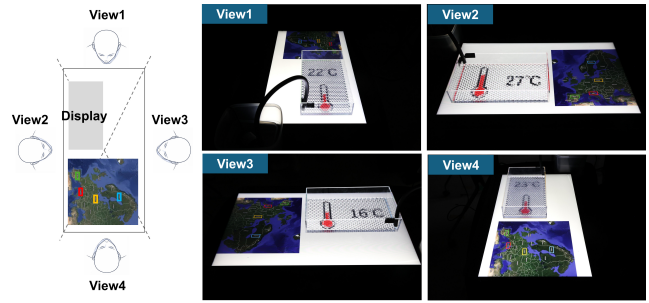


Figure 3: Example application showing regional temperature information revealed from different viewing perspectives.

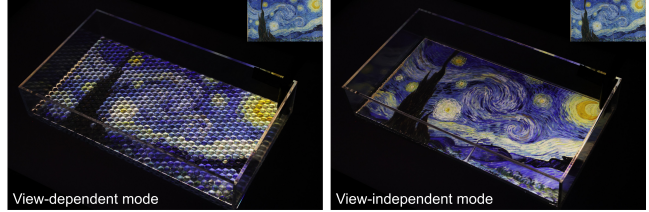


Figure 4: Comparison of the same image displayed in view-dependent mode (left) and view-independent mode (right).

the lens array and the intended viewing directions, and shown on the flat-panel display.

3 Application

Figure 3 shows an example application of LentiMorpho operating in the view-dependent mode. In this mode, the displayed content changes according to the viewing direction, allowing viewers moving around the display to observe different images with minimal crosstalk. Figure 4 also presents a comparison of display quality between the view-dependent mode and the view-independent mode.

In the view-independent mode, LentiMorpho presents identical content uniformly across all viewing directions while preserving the image quality of the underlying flat-panel display. This example demonstrates LentiMorpho alleviates the trade-off between the number of supported viewing directions and image quality inherent

in conventional view-dependent displays. In this way, LentiMorpho provides a technical option for extending existing view-independent displays with view-dependent presentation capabilities.

References

- Yasuaki Kakehi, Makoto Iida, Takeshi Naemura, Yoshinari. Shirai, Mitsunori Matsushita, and Takeshi Ohguro. 2005. Lumisight Table: an interactive view-dependent tabletop display. *IEEE Computer Graphics and Applications* 25, 1 (2005), 48–53. doi:10.1109/MCG.2005.14
- Douglas Lanman, Matthew Hirsch, Yunhee Kim, and Ramesh Raskar. 2010. Content-adaptive parallax barriers: optimizing dual-layer 3D displays using low-rank light field factorization. In *ACM SIGGRAPH Asia 2010 Papers* (Seoul, South Korea) (SIGGRAPH ASIA '10). Association for Computing Machinery, New York, NY, USA, Article 163, 10 pages. doi:10.1145/1866158.1866164
- Marc Levoy and Pat Hanrahan. 1996. Light field rendering. In *Proceedings of the 23rd Annual Conference on Computer Graphics and Interactive Techniques (SIGGRAPH '96)*. Association for Computing Machinery, New York, NY, USA, 31–42. doi:10.1145/237170.237199
- Yasuhiro Takaki and Nichiyo Nago. 2010. Multi-projection of lenticular displays to construct a 256-view super multi-view display. *Opt. Express* 18, 9 (Apr 2010), 8824–8835. doi:10.1364/OE.18.008824